VU TUAN HUNG

Machine Learning Engineer

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EDUCATION

HCMC University of Technology (HCMUT) Sep. 2020 – Apr. 2025

Bachelor of Engineering in Electronics & Telecommunications

TOEIC : 870

WORK EXPERIENCES

Puwell Cloud Technology | *AI Engineer* Mar. 2025 – Jun. 2026

Main Responsibilities:

- Built AI features for cameras by researching data and solutions, designed complete end-to-end pipeline on chips.
- Improved model on-device performance, minimizing size and latency with minimal accuracy loss.
- Planned test scenarios and ran experiments/evaluations to ensure target metrics.
- Collaborated with embedded team for real-world validation.

Delivered Projects:

- **Human Detection:** Built a real-time human detection system for cameras by optimizing MobileNet classifier and designing a custom Multi-ROI algorithm, deploying full pipeline on ESP32.
  Achievement (on-device): Optimized model latency to real-time (<50ms/frame), increased full pipeline's sensitivity to $>$80% while driving FAR down to $<$20%
- **Multiclass Object Detection (person, animals, vehicle, package):** Built multi-class object detection feature for smart home cameras by developing YOLOv5n and YOLOv5-Lite; optimized and deployed models on two embedded platforms: Ingenic T32 and Allwinner NPU.
  Achievement: Met target metrics for all classes; **shipped the feature to production on Virtavo cameras**.
- **Audio Denoising:** Built an audio enhancement for cameras by customizing DeepFilterNet-based denoising pipeline, optimized for the Allwinner NPU.
  Achievement (on-device): Boosted voice clarity metrics to PESQ 3.2 and STOI 1.0.

QBOX | *Edge AI Intern* Jun. 2023 – Dec. 2023

Project: Automatic shoe insole counting machine

- Built a real-time shoe insole counting system for manufacturing by optimizing YOLOv8 segmentation model with custom image processing algorithms, fully deployed on the Jetson Nano.

PERSONAL PROJECT

Biometric Timekeeping System (*Graduation Thesis*) | **Demo:** github.com/vutuanhungkkk/Timekeeping_machine

- Developed an employee management system with fingerprint recognition and face recognition/anti-spoofing, optimized for edge device (Jetson Nano). Integrated Flask web with SQL database for user management and timekeeping.

SKILLS SUMMARY

Personal strengths : Fluent in English; proficient in self-study, research and teamwork.

Technical : Data processing, deep learning model development & optimization (train, fine-tune, architecture modification, cross-framework conversion, quantization), edge deployment.

Hardware : ESP32, Jetson Nano, Allwinner NPU, Ingenic T32

Frameworks & Libraries : PyTorch, TensorFlow, OpenCV, NumPy, ONNX, TensorRT, TFLite, Espdl

Programming Languages : Python, C/C++, Linux Shell

Tools : Git, Google Colab, Kaggle, VSCode, TensorBoard, ComfyUI

Basic Knowledge : Reinforcement Learning, LLM API calling, RAG pipeline design, vector databases (FAISS)